OLA Ride Booking Analysis

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Executive Summary

Summary

The **OLA Ride Booking Analysis** project is an end-to-end Business Intelligence solution developed using **Microsoft Excel, SQL, and Power BI**. The primary objective of this project is to transform raw ride booking data into meaningful business insights that support operational efficiency and data-driven decision-making.

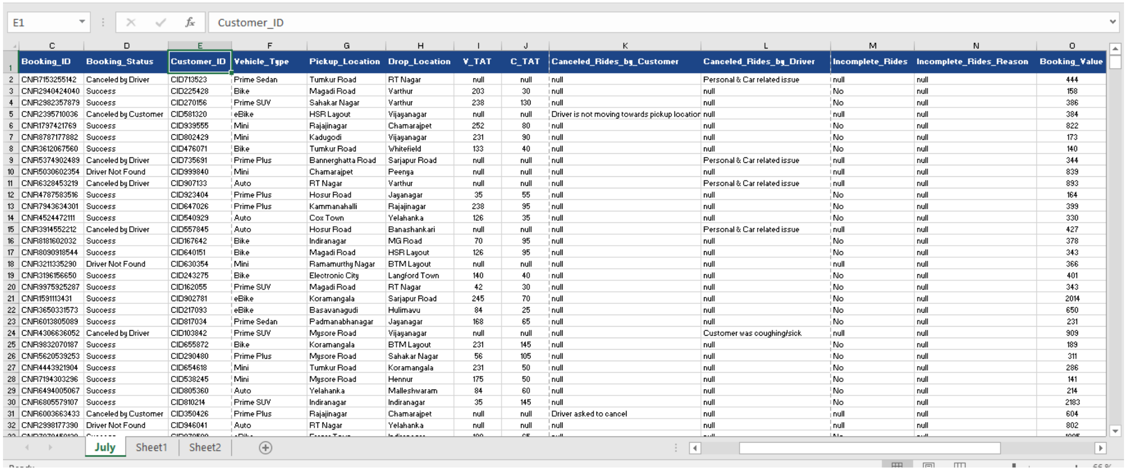
The analysis is based on a dataset containing approximately **40,000 ride booking records**, covering booking status, vehicle categories, ride distance, booking value, payment methods, and customer feedback. The project demonstrates the complete analytics workflow, including data cleaning, SQL-based querying, interactive dashboard development, KPI tracking, and business insight generation.

The Executive Dashboard provides a consolidated view of key performance indicators, enabling stakeholders to monitor booking trends, revenue, ride success rates, customer satisfaction, and overall business performance through interactive visualizations.

Project Objectives

- Analyze ride booking performance and operational trends.
- Monitor key business KPIs through interactive dashboards.
- Evaluate vehicle-wise performance and revenue contribution.
- Identify booking and cancellation patterns.

Dataset Description



E1 Customer_ID

	Booking_ID	Booking_Status	Customer_ID	Vehicle_Type	Pickup_Location	Drop_Location	V_TAT	C_TAT	Canceled_Rides_by_Customer	Canceled_Rides_by_Driver	Incomplete_Rides	Incomplete_Rides_Reason	Booking_Value
2	CNR7153255142	Canceled by Driver	CID713523	Prime Sedan	Tumkur Road	RT Nagar	null	null	null	Personal & Car related issue	null	null	444
3	CNR2940424040	Success	CID225428	Bike	Magadi Road	Varthur	203	30	null	null	No	null	158
4	CNR2982357879	Success	CID270156	Prime SUV	Sahakar Nagar	Varthur	238	130	null	null	No	null	386
5	CNR2395710036	Canceled by Customer	CID581320	eBike	HSR Layout	Vijayanagar	null	null	Driver is not moving towards pickup locatior	null	null	null	384
6	CNR1797421769	Success	CID939555	Mini	Rajajinagar	Chamarajpet	252	80	null	null	No	null	822
7	CNR8787177882	Success	CID802429	Mini	Kadugodi	Vijayanagar	231	90	null	null	No	null	173
8	CNR3612067560	Success	CID476071	Bike	Tumkur Road	Whitefield	133	40	null	null	No	null	140
9	CNR5374902489	Canceled by Driver	CID735691	Prime Plus	Bannerghatta Road	Sarjapur Road	null	null	null	Personal & Car related issue	null	null	344
10	CNR5030602354	Driver Not Found	CID999840	Mini	Chamarajpet	Peenya	null	null	null	null	null	null	839
11	CNR6328453219	Canceled by Driver	CID907133	Auto	RT Nagar	Varthur	null	null	null	Personal & Car related issue	null	null	893
12	CNR4787583516	Success	CID923404	Prime Plus	Hosur Road	Jayanagar	35	55	null	null	No	null	164
13	CNR7943634301	Success	CID647026	Prime Plus	Kammanahalli	Rajajinagar	238	95	null	null	No	null	399
14	CNR4524472111	Success	CID540929	Auto	Cox Town	Yelahanka	126	35	null	null	No	null	330
15	CNR3914552212	Canceled by Driver	CID557845	Auto	Hosur Road	Banashankari	null	null	null	Personal & Car related issue	null	null	427
16	CNR8181602032	Success	CID167642	Bike	Indiranagar	MG Road	70	95	null	null	No	null	378
17	CNR8090918544	Success	CID640151	Bike	Magadi Road	HSR Layout	126	95	null	null	No	null	343
18	CNR3211335290	Driver Not Found	CID630354	Mini	Ramamurthy Nagar	BTM Layout	null	null	null	null	null	null	366
19	CNR3196156650	Success	CID243275	Bike	Electronic City	Langford Town	140	40	null	null	No	null	401
20	CNR9975925287	Success	CID162055	Prime SUV	Magadi Road	RT Nagar	42	30	null	null	No	null	343
21	CNR1591113431	Success	CID902781	eBike	Koramangala	Sarjapur Road	245	70	null	null	No	null	2014
22	CNR3650331573	Success	CID217093	eBike	Basavanagudi	Hulimavu	84	25	null	null	No	null	650
23	CNR6013805089	Success	CID817034	Prime Sedan	Padmanabhanagar	Jayanagar	168	65	null	null	No	null	231
24	CNR4306636052	Canceled by Driver	CID103842	Prime SUV	Mysore Road	Vijayanagar	null	null	null	Customer was coughing/sick	null	null	909
25	CNR9832070187	Success	CID655872	Bike	Koramangala	BTM Layout	231	145	null	null	No	null	189
26	CNR5620539253	Success	CID290480	Prime Plus	Mysore Road	Sahakar Nagar	56	105	null	null	No	null	311
27	CNR4443921904	Success	CID654618	Mini	Tumkur Road	Koramangala	231	50	null	null	No	null	286
28	CNR7194303296	Success	CID538245	Mini	Mysore Road	Hennur	175	50	null	null	No	null	141
29	CNR6494005067	Success	CID805360	Auto	Yelahanka	Malleshwaram	84	60	null	null	No	null	214
30	CNR6805579107	Success	CID810214	Prime SUV	Indiranagar	Indiranagar	35	145	null	null	No	null	2183
31	CNR6003663433	Canceled by Customer	CID350426	Prime Plus	Rajajinagar	Chamarajpet	null	null	Driver asked to cancel	null	null	null	604
32	CNR2998177390	Driver Not Found	CID946041	Auto	RT Nagar	Yelahanka	null	null	null	null	null	null	802

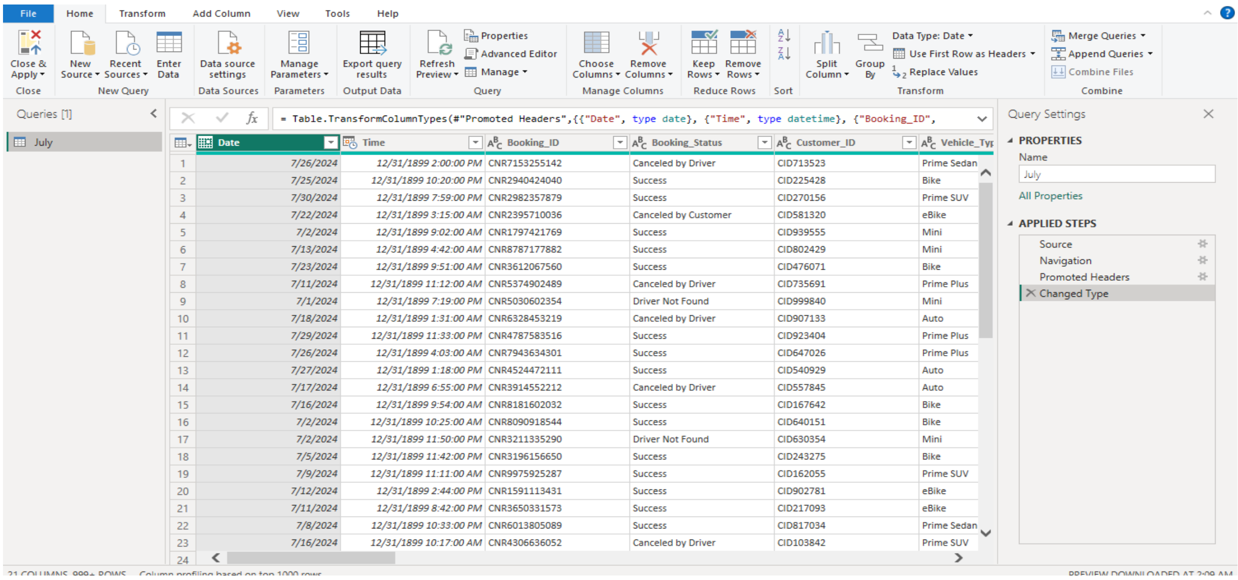
July Sheet1 Sheet2

Dataset Overview :

The dataset contains booking details, vehicle type, booking status, booking value, ride distance, payment method, customer rating, driver rating.

Attribute	Details
Dataset Name	OLA Ride Booking Dataset
Total Records	40,000
File Format	Microsoft Excel
Number of Columns	20
Tools Used	Excel, SQL, Power BI
Domain	Transportation / Ride Booking

Data Cleaning & Power Query



Summary

Before analysis, the OLA ride booking dataset was cleaned and transformed using **Power Query** in Microsoft Power BI. The data preparation process ensured consistency, accuracy, and reliability for SQL analysis and dashboard development. Data types were validated, column headers were standardized, and the dataset was reviewed for missing values and duplicates. These transformations created a clean and structured dataset suitable for business intelligence reporting.

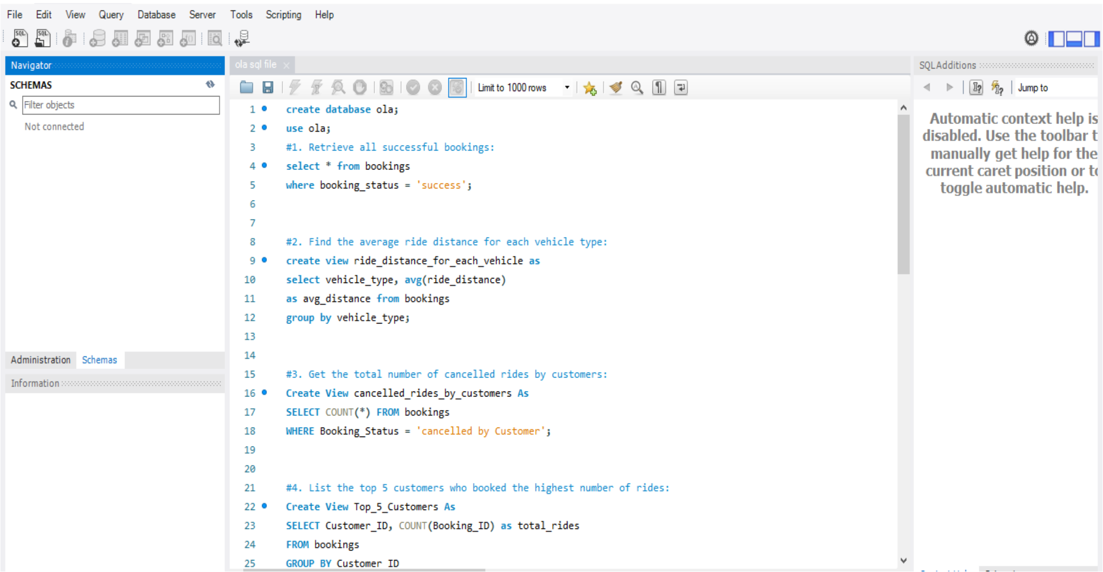
Key Data Preparation Steps

- Imported the dataset into Power BI.
- Verified and corrected data types.
- Standardized column headers and formats.
- Reviewed missing and duplicate records.

Key Takeaways

- Improved data quality and consistency.
- Streamlined the ETL process using Power Query.
- Prepared reliable data for SQL analysis and dashboard visualization.

SQL Analysis



SQL was used to import the OLA ride booking dataset into a relational database and perform business-focused analysis. Various SQL queries were executed to retrieve meaningful insights, calculate key performance metrics, and analyze booking trends. These query results served as the foundation for building interactive Power BI dashboards.

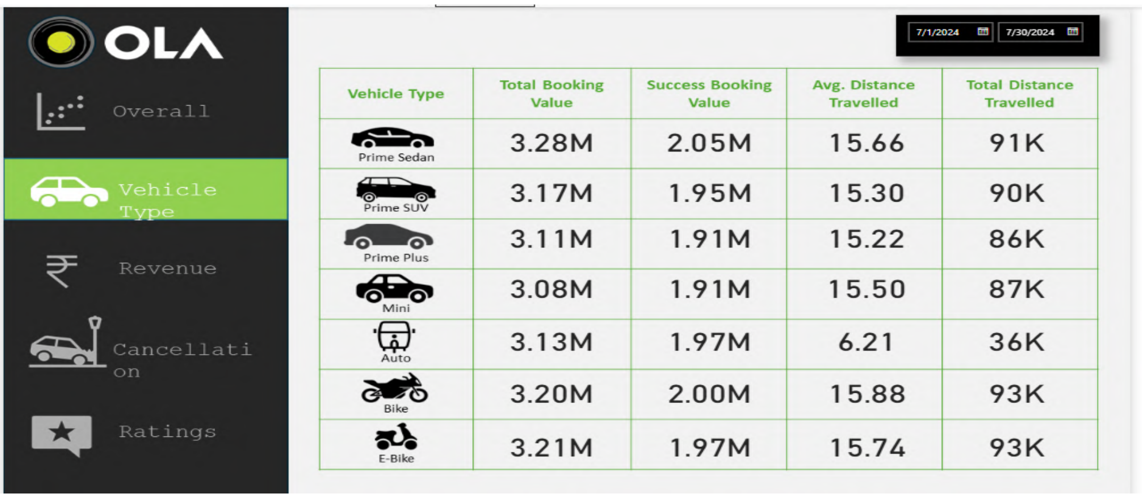
SQL Operations Performed

- Created the database and imported the dataset.
- Retrieved booking records using SQL queries.
- Performed data aggregation using **COUNT(), SUM(),** and **AVG()**.
- Applied **WHERE**, **GROUP BY**, and **ORDER BY** clauses for data filtering and analysis.
- Generated business insights for dashboard development.

Business Questions Answered

- What is the total number of bookings?
- Which vehicle type has the highest bookings?
- What is the total booking revenue?
- What are the booking cancellation trends?
- How do customer and driver ratings compare?

Vehicle Analysis



Vehicle Type	Total Booking Value	Success Booking Value	Avg. Distance Travelled	Total Distance Travelled
Prime Sedan	3.28M	2.05M	15.66	91K
Prime SUV	3.17M	1.95M	15.30	90K
Prime Plus	3.11M	1.91M	15.22	86K
Mini	3.08M	1.91M	15.50	87K
Auto	3.13M	1.97M	6.21	36K
Bike	3.20M	2.00M	15.88	93K
E-Bike	3.21M	1.97M	15.74	93K

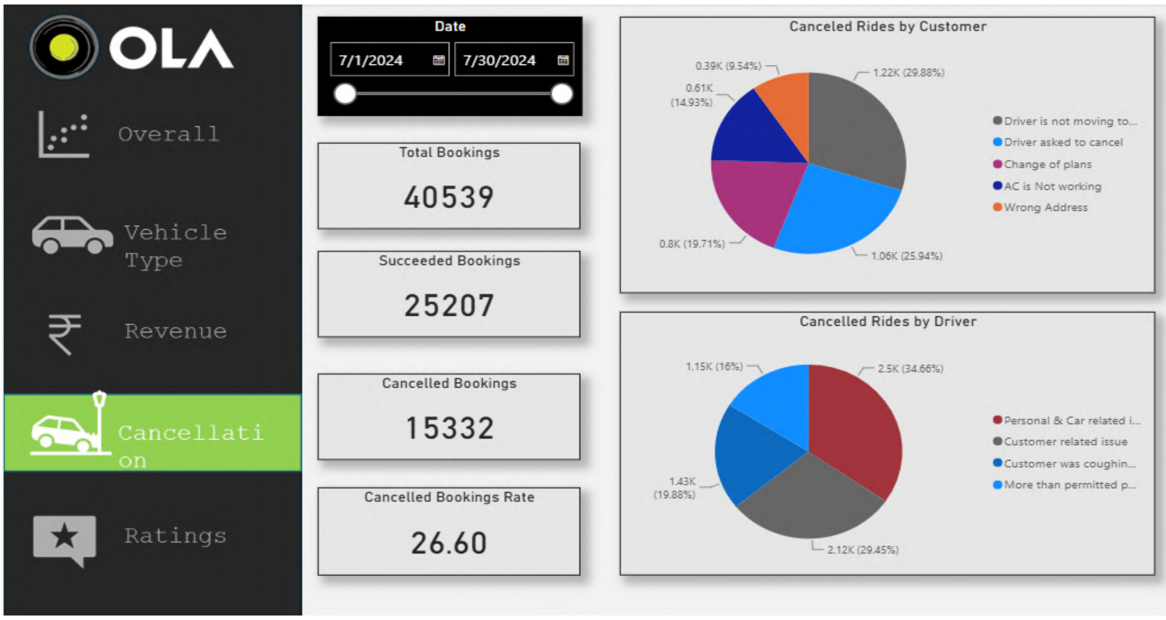
Summary

This dashboard analyzes the performance of different vehicle categories by comparing booking volume, revenue contribution, and customer demand. It helps identify the most preferred vehicle types and supports data-driven decisions for fleet management and resource allocation. The analysis provides valuable insights into customer preferences and operational performance across various ride categories.

Key Metrics Analyzed

- Total Bookings by Vehicle Type
- Booking Revenue by Vehicle Category
- Average Ride Distance
- Customer Demand Distribution
- Vehicle-wise Performance Comparison

Cancellation & Ratings



Summary

This dashboard analyzes ride cancellation patterns to understand the factors affecting booking completion and service efficiency. It highlights customer-initiated and driver-initiated cancellations, enabling the identification of operational challenges and opportunities to improve ride success rates. Monitoring cancellation trends helps enhance customer satisfaction, optimize driver availability, and improve overall platform performance.

Key Metrics Analyzed

- Customer Cancellation Rate
- Driver Cancellation Rate
- Successful Ride Percentage
- Cancellation Reasons
- Booking Status Distribution

Business Insights

Summary

The analysis of the OLA ride booking dataset provided valuable insights into booking performance, customer behavior, revenue trends, and operational efficiency. By integrating SQL analysis with Power BI visualizations, key performance indicators were monitored to identify trends and support data-driven decision-making. These insights can help optimize fleet operations, improve customer satisfaction, and enhance overall business performance.

Key Business Insights

- Successful rides contributed the highest share of total bookings, indicating strong operational performance.
- Certain vehicle categories consistently generated higher booking demand and revenue.
- Ride cancellations impacted overall booking success and highlighted opportunities for operational improvement.
- Customer and driver ratings reflected overall service quality and user satisfaction.
- Interactive dashboards enabled quick monitoring of KPIs and business performance.

Business Impact

The insights generated from this analysis can support better fleet allocation, reduce ride cancellations, improve customer experience, and enable management to make informed business decisions using real-time performance metrics.

Business Recommendations & Future Scope

Summary

Based on the insights obtained from the OLA Ride Booking Analysis, several recommendations can be made to improve operational efficiency, customer satisfaction, and overall business performance. These suggestions are derived from booking trends, vehicle performance, cancellation patterns, and customer feedback. Implementing these recommendations can help optimize business operations and support long-term growth.

Business Recommendations

- Optimize vehicle allocation based on demand to reduce customer waiting time.
- Implement strategies to minimize ride cancellations through improved driver availability and customer communication.
- Focus on high-performing vehicle categories to maximize revenue and operational efficiency.
- Continuously monitor key performance indicators (KPIs) to support data-driven decision-making.
- Enhance customer experience by improving service quality and addressing feedback from customer ratings.

Future Scope

- Integrate real-time booking data for live dashboard monitoring.
- Develop predictive models to forecast ride demand and peak booking periods.
- Perform geographical analysis to identify high-demand service locations.
- Build advanced dashboards with automated data refresh and scheduled reporting.
- Expand the analysis by incorporating seasonal trends and customer segmentation.

Conclusion

The **OLA Ride Booking Analysis** project successfully demonstrates an end-to-end Business Intelligence workflow using **Microsoft Excel, SQL, and Power BI**. Starting with raw ride booking data, the project involved data cleaning, SQL-based analysis, interactive dashboard development, and business insight generation to support data-driven decision-making.

Through comprehensive analysis of booking trends, vehicle performance, revenue, ride cancellations, and customer feedback, the project transformed raw operational data into meaningful insights. The interactive dashboards enable stakeholders to monitor key performance indicators (KPIs), evaluate business performance, and identify opportunities for operational improvement.

This project highlights practical skills in **data preparation, SQL querying, data visualization, dashboard design, KPI analysis, and business storytelling**. It reflects the ability to solve real-world business problems using data analytics and Business Intelligence tools, making it a valuable addition to a professional portfolio.

Skills Demonstrated

- Microsoft Excel
- SQL (Database Management & Querying)
- Power BI
- Power Query
- Data Cleaning & Transformation
- Data Visualization
- Dashboard Development
- Business Intelligence
- KPI Analysis
- Analytical Thinking
- Data Storytelling

Project Outcome

- Developed an interactive Business Intelligence dashboard.
- Converted raw booking data into actionable business insights.
- Identified key trends in bookings, revenue, and cancellations.
- Demonstrated practical application of SQL and Power BI for business analytics.
- Created a scalable reporting solution to support informed decision-making.